N5000 - Dielectric Properties Table
AMMA Manufacturing Location

LAMINATE (Available from AGC U.S. manufacturing location – AMMA):

Thickness (inch)	&	Tol.	Construction					RC%	Nominal Thickness (inch)	Nominal Thickness (µm)	1 MHz Dk	1 GHz Dk	2.5 GHz Dk	10 GHz Dk
0.0020	±	0.0005	1	106				63%	0.0017	42	3.60	3.46	3.41	3.38
0.0027	±	0.0005	1	1080				57%	0.0027	68	3.73	3.60	3.50	3.47
0.0030	±	0.0005	1	1080				63%	0.0032	81	3.60	3.47	3.41	3.38
0.0035	±	0.0005	2	106				63%	0.0033	85	3.60	3.46	3.41	3.38
0.0040	±	0.0005	1	106	+	1	1080	56%	0.0040	102	3.75	3.61	3.51	3.48
0.0040	±	0.0005	1	2116				46%	0.0043	110	4.05	3.92	3.71	3.70
0.0045	±	0.0005	1	2116				47%	0.0045	115	4.01	3.87	3.68	3.66
0.0050	±	0.0007	1	1080	+	1	1080	55%	0.0050	128	3.79	3.65	3.54	3.51
0.0050	±	0.0007	1	2113	+	1	106	53%	0.0053	135	3.82	3.69	3.56	3.53
0.0060	±	0.0007	1	1080	+	1	2113	53%	0.0063	160	3.84	3.71	3.57	3.55
0.0060	±	0.0007	1	2116	+	1	106	51%	0.0062	157	3.89	3.76	3.60	3.58
0.0070	±	0.0010	2	2113				50%	0.0073	184	3.93	3.80	3.63	3.61
0.0080	±	0.0010	2	1080	+	1	2113	54%	0.0090	229	3.81	3.68	3.55	3.52
0.0080	±	0.0010	1	2116	+	1	2113	48%	0.0081	207	3.97	3.84	3.66	3.64
0.0090	±	0.0010	2	2116				47%	0.0090	230	4.01	3.87	3.68	3.66
0.0100	±	0.0010	2	1080	+	1	2116	56%	0.0109	277	3.76	3.63	3.52	3.49
0.0100	±	0.0010	2	2116	+	1	106	49%	0.0107	272	3.94	3.80	3.64	3.61
0.0120	±	0.0015	2	1080	+	1	7628	46%	0.0126	319	4.04	3.90	3.70	3.69
0.0140	±	0.0015	2	2113	+	1	7628	44%	0.0144	367	4.11	3.97	3.75	3.73
0.0160	±	0.0015	2	2116	+	1	7628	43%	0.0162	412	4.13	3.99	3.76	3.75

N5000 - Dielectric Properties Table
AMMA Manufacturing Location

Laminate (continued):

Thickness (inch)	&	Tol.	Construction					RC%	Nominal Thickness (inch)	Nominal Thickness (µm)	1 MHz Dk	1 GHz Dk	2.5 GHz Dk	10 GHz Dk
0.0180	±	0.0015	2	7628	+	1	2116	41%	0.0189	480	4.22	4.08	3.82	3.81
0.0200	±	0.0025	2	2113	+	2	7628	42%	0.0216	549	4.17	4.03	3.79	3.77
0.0210	±	0.0025	3	7628				39%	0.0216	547	4.29	4.15	3.87	3.86
0.0240	±	0.0025	3	2113	+	2	7628	43%	0.0253	641	4.13	3.99	3.76	3.75
0.0280	±	0.0025	2	2116	+	3	7628	41%	0.0302	768	4.22	4.08	3.82	3.81
0.0394	±	0.0040	2	1080	+	5	7628	40%	0.0406	1031	4.24	4.10	3.84	3.82
0.0440	±	0.0050	6	7628	+	1	2113	39%	0.0467	1187	4.26	4.12	3.85	3.84
0.0590	±	0.0050	2	1080	+	8	7628	40%	0.0629	1596	4.24	4.10	3.84	3.82

PREPREG (Available from AGC U.S. manufacturing location - AMMA):

Glass	RC%	Thickness (inches)	Thickness (µm)	1 MHz Dk	1 GHz Dk	2.5 GHz Dk	10 GHz Dk
106	76%	0.0028	71	3.32	3.19	3.22	3.18
1080	65%	0.0035	88	3.55	3.41	3.37	3.34
2113	58%	0.0046	117	3.71	3.57	3.48	3.45
2116	54%	0.0054	138	3.81	3.68	3.55	3.52
7628	44%	0.0082	208	4.10	3.97	3.74	3.73

N5000 - Dielectric Properties Table
AMMS Manufacturing Location

LAMINATE (Available from AGC Singapore manufacturing location – AMMS):

Thickness (inch)	&	Tol.	Construction					RC%	Nominal Thickness (inch)	Nominal Thickness (µm)	1 MHz Dk	1 GHz Dk	2.5 GHz Dk	10 GHz Dk
0.0020	±	0.0005	1	106				68%	0.0019	49	3.48	3.35	3.33	3.29
0.0027	±	0.0005	1	1080				57%	0.0026	66	3.73	3.60	3.50	3.47
0.0030	±	0.0005	1	1080				63%	0.0032	80	3.59	3.46	3.40	3.37
0.0035	±	0.0005	2	106				63%	0.0033	83	3.59	3.46	3.40	3.37
0.0040	±	0.0005	1	106	+	1	1080	56%	0.0039	99	3.75	3.61	3.51	3.48
0.0040	±	0.0005	1	2116				45%	0.0041	104	4.09	3.95	3.73	3.72
0.0045	±	0.0005	1	2116				48%	0.0044	113	3.99	3.86	3.67	3.65
0.0050	±	0.0007	1	1080	+	1	1080	55%	0.0049	124	3.79	3.66	3.54	3.51
0.0050	±	0.0007	1	2113	+	1	106	54%	0.0052	132	3.81	3.68	3.55	3.53
0.0060	±	0.0007	1	1080	+	1	2113	53%	0.0062	157	3.84	3.71	3.57	3.55
0.0060	±	0.0007	1	2116	+	1	106	51%	0.0061	154	3.88	3.75	3.60	3.57
0.0070	±	0.0010	2	2113				50%	0.0072	182	3.92	3.78	3.62	3.60
0.0080	±	0.0010	2	1080	+	1	2113	54%	0.0088	224	3.81	3.67	3.55	3.52
0.0080	±	0.0010	1	2116	+	1	2113	49%	0.0080	204	3.96	3.82	3.65	3.63
0.0090	±	0.0010	2	2116				48%	0.0089	225	3.99	3.86	3.67	3.65
0.0100	±	0.0010	2	1080	+	1	2116	56%	0.0107	273	3.75	3.62	3.51	3.48
0.0100	±	0.0010	2	2116	+	1	106	50%	0.0105	267	3.93	3.79	3.63	3.61
0.0120	±	0.0015	3	2116				45%	0.0123	313	4.09	3.95	3.73	3.72
0.0140	±	0.0015	4	2113				47%	0.0133	337	4.01	3.87	3.68	3.66
0.0150	±	0.0015	2	2116	+	2	2113	46%	0.0148	377	4.05	3.92	3.71	3.69

N5000 - Dielectric Properties Table
AMMS Manufacturing Location

Laminate (Continued) (Available from AGC Singapore manufacturing location - AMMS):

Thickness (inch)	&	Tol.	Construction					RC%	Nominal Thickness (inch)	Nominal Thickness (µm)	1 MHz Dk	1 GHz Dk	2.5 GHz Dk	10 GHz Dk
0.0160	±	0.0015	4	2116				45%	0.0164	417	4.09	3.95	3.73	3.72
0.0180	±	0.0015	4	2116	+	1	106	47%	0.0184	466	4.02	3.88	3.69	3.67
0.0200	±	0.0025	5	2116				45%	0.0205	521	4.09	3.95	3.73	3.72
0.0210	±	0.0025	2	2116	+	3	2116	46%	0.0212	538	4.05	3.91	3.71	3.69
0.0240	±	0.0025	6	2116				45%	0.0246	626	4.09	3.95	3.73	3.72
0.0280	±	0.0025	7	2116				45%	0.0287	730	4.09	3.95	3.73	3.72
0.0290	±	0.0025	4	2116	+	3	2116	46%	0.0297	755	4.04	3.91	3.71	3.69
0.0370	±	0.0040	2	2116	+	7	2116	45%	0.0376	955	4.06	3.93	3.72	3.70
0.0400	±	0.0040	10	2116				45%	0.0411	1043	4.09	3.95	3.73	3.72
0.0440	±	0.0050	11	2116				45%	0.0452	1147	4.09	3.95	3.73	3.72
0.0490	±	0.0050	12	2116				45%	0.0493	1251	4.09	3.95	3.73	3.72
0.0570	±	0.0050	2	2116	+	12	2116	45%	0.0581	1477	4.07	3.94	3.72	3.71
0.0600	±	0.0050	15	2116				45%	0.0616	1564	4.09	3.95	3.73	3.72
0.0620	±	0.0050	2	2116	+	13	2116	45%	0.0622	1581	4.07	3.94	3.72	3.71

N5000 - Dielectric Properties Table
AMMS Manufacturing Location

PREPREG (Available from AGC Singapore manufacturing location - AMMS):

Glass	RC%	Thickness (inches)	Thickness (µm)	1 MHz Dk	1 GHz Dk	2.5 GHz Dk	10 GHz Dk
106	76%	0.0027	69	3.32	3.19	3.22	3.18
1080	65%	0.0034	86	3.55	3.41	3.37	3.34
2113	58%	0.0045	114	3.71	3.57	3.48	3.45
2116	45%	0.0042	106	4.07	3.93	3.72	3.71
2116	48%	0.0045	114	3.98	3.84	3.66	3.64
2116	54%	0.0053	134	3.81	3.68	3.55	3.52

These laminate and prepreg items are representative of available materials. These items and their associated values may not be regularly available and additional lead times or minimum order quantities could apply. Please contact our office or your local technical representative for assistance with choosing the best material for your design and application.